The Chain Rule (14.5)

$f = f(x,y,z)$

$$\boxed{\Delta f \approx \underbrace{f_x}_{\frac{\partial f}{\partial x}(x,y,z)} \Delta x + f_y \Delta y + f_z \Delta z}$$

$x = x(t) \quad y = y(t) \quad z = z(t)$

$$\Delta f \approx f_x \underbrace{\frac{dx}{dt} \Delta t}_{\approx \Delta x} + f_y \frac{dy}{dt} \Delta t + f_z \frac{dz}{dt} \Delta t$$

Chain rule

$$\boxed{\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}}$$

Example: $f = x^2 y + z$

$x = t \quad y = e^t \quad z = \sin t$

$$\frac{df}{dt} = 2xy \cdot 1 + x^2 \cdot e^t + 1 \cdot \cos(t)$$

$$= 2t e^t + t^2 e^t + \cos(t) //$$

Note: To know $\left.\frac{df}{dt}\right|_{t=1}$, need:

- f_x, f_y, f_z for $(x,y,z) = (1, e^1, \sin 1)$
- $\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}$ for $t = 1$.

Application: Product & quotient rules

$f = uv \qquad u = u(t), \quad v = v(t)$

$$\frac{d(uv)}{dt} = f_u \frac{du}{dt} + f_v \frac{dv}{dt}$$

$$= v \frac{du}{dt} + u \frac{dv}{dt}$$

$g = u/v \qquad u = u(t), \quad v = v(t)$

$$\frac{d(u/v)}{dt} = \frac{1}{v} \frac{du}{dt} + \left(\frac{-u}{v^2}\right) \frac{dv}{dt}$$

Chain rule w/ more variables

$f = f(x,y) \qquad x = x(u,v)$

$y = y(u,v)$

$$\frac{\partial f}{\partial u} = f_x \frac{\partial x}{\partial u} + f_y \frac{\partial y}{\partial u}$$

$$\frac{\partial f}{\partial v} = f_x \frac{\partial x}{\partial v} + f_y \frac{\partial y}{\partial v}$$

Example: $x = r\cos\theta$

$y = r\sin\theta$

$$f_r = f_x \frac{\partial x}{\partial r} + f_y \frac{\partial y}{\partial r}$$

$$= f_x \cos\theta + f_y \sin\theta$$

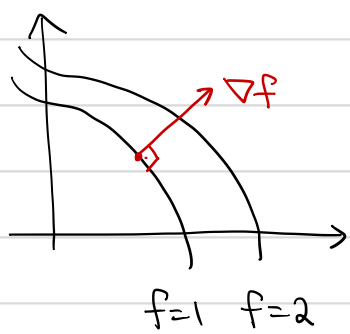
$$f_\theta = f_x \frac{\partial x}{\partial \theta} + f_y \frac{\partial y}{\partial \theta}$$

$$= f_x(-r\sin\theta) + f_y r\cos\theta$$

The Gradient Vector (14.6)

$$\boxed{\nabla f = \langle f_x, f_y, f_z \rangle}$$

Theorem: $\nabla f \perp$ level set $\{f = \text{constant}\}$



Example: $f = ax + by + cz$

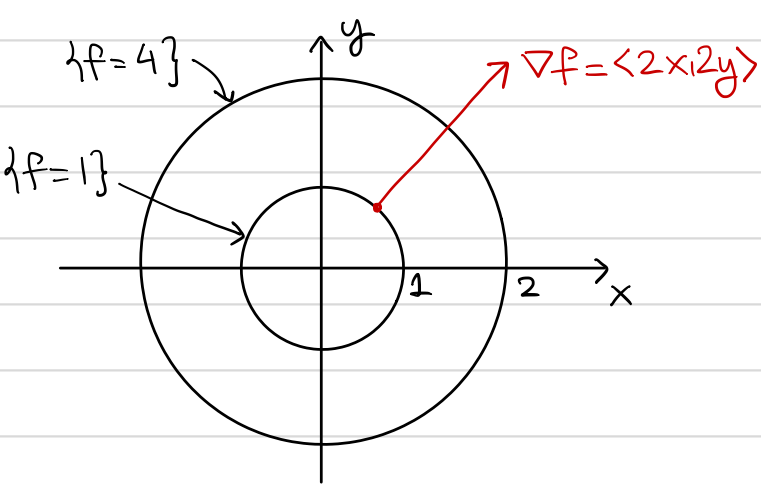
Level surface $\{\overbrace{ax+by+cz}^{\langle a,b,c\rangle \cdot \langle x,y,z\rangle} = d\}$ is plane with normal

$$\nabla f = \langle a, b, c \rangle.$$

$$\uparrow$$

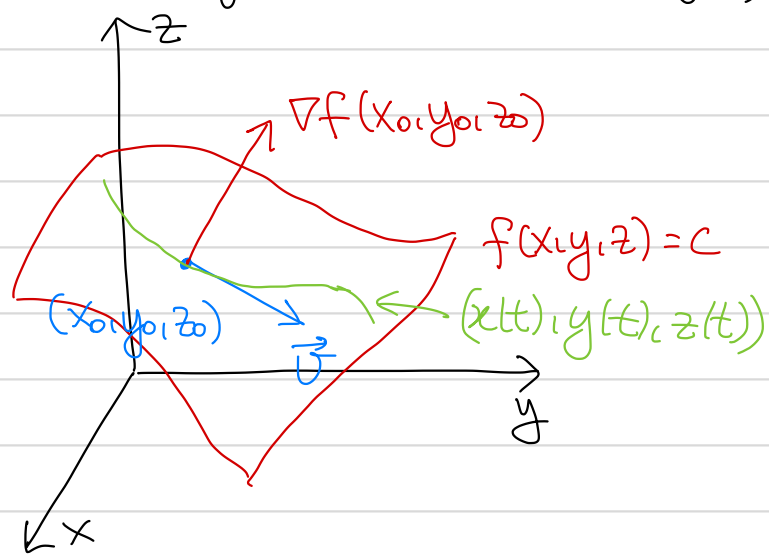
$$\frac{\partial f}{\partial x}$$

Example: $f(x,y) = x^2 + y^2$



Proof: Let:

- $S := \{(x,y,z) \in \mathbb{R}^3 \mid f(x,y,z) = c\}$
- $\vec{v}$ tangent to S at (x_0, y_0, z_0).



By definition of "tangent":

$$\vec{v} = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle \Big|_{t=0}$$

for some curve

$$(x(t), y(t), z(t)), \quad t \in \mathbb{R}$$

contained in S.

means: $f(x(t), y(t), z(t)) = c \ \forall t$

Therefore:

$$0 = \frac{df}{dt} = f_x \cdot \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}$$

$$= \langle f_x, f_y, f_z \rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle$$

$$= \nabla f \cdot \vec{v} \implies \nabla f \perp \vec{v}. \quad \blacksquare$$

Application: Find tangent plane of the surface

$$x^2+y^2-z^2=4$$

at $(2,1,1)$.

Level set $f=4$, where

$$f=x^2+y^2-z^2.$$

Gradient: ← normal

$$\nabla f = \langle 2x, 2y, -2z\rangle$$
$$= \langle 4, 2, -2\rangle$$

Tangent plane: ← contains $(2,1,1)$

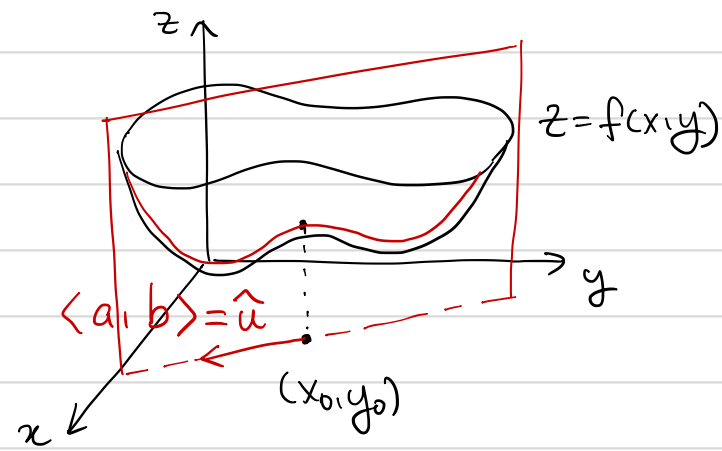
$$4x+2y-2z=8$$

Directional derivatives

$f=f(x,y) \rightsquigarrow \dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial y}$

Direction: $\hat{i}$, $\hat{j}$

What if I move in another direction $\hat{u}$ = unit vector?



Def: $D_{\hat{u}}f(x_0,y_0) =$

$$= \frac{d}{dt} f(x_0+at, y_0+bt)\Big|_{t=0}$$

"Directional derivative"

$D_{\hat{u}}f$ = slope of slice of graph by vertical plane $\parallel \hat{u}$

$$D_{\hat{u}}f = \frac{df}{dt} = f_x \overbrace{\frac{dx}{dt}}^{a} + f_y \overbrace{\frac{dy}{dt}}^{b}$$

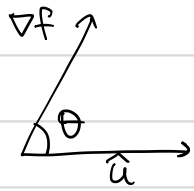
$$= \nabla f \cdot \hat{u}$$

$\Rightarrow D_{\hat{u}}f$ = component of ∇f in direction of $\hat{u}$

Example: $D_{\hat{i}}f = \nabla f \cdot \hat{i} = \dfrac{\partial f}{\partial x}$

Meaning of the gradient

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$
$$= |\nabla f| \cdot |\hat{u}| \cdot \cos\theta$$
$$= |\nabla f| \cdot \cos\theta$$



$\rightarrow$ max (≥ 0) when $\theta = 0$
min (≤ 0) when $\theta = \pi$
0 when $\theta = \pi/2$

Direction of ∇f =
= direction along which
f increases the fastest.

Magnitude $|\nabla f|$ =
= Fastest rate of
increase.